

Quant Research Scout

Daily report - 2026-05-11

5 findings saved this run

Digest

Daily scan 2026-05-11: 5 strong findings today across regime detection and VaR control.

1. Adaptive VaR Control for Standardized Option Books under Marking Frictions

<https://arxiv.org/abs/2604.0003>

Source: arxiv Type: model Relevance: 0.85 Replicability: ***-- (3/5)

Authors: Tengan Zhong

Published: 2026-04-03

TLDR

(not provided)

Summary

Develops a tail-risk framework for standardized option books that combines a base conditional-quantile forecast with sequential conformal recalibration. Tested on SPX option data 2018-2025; the uncalibrated base systematically underestimates downside risk, and sequential recalibration restores coverage and improves rolling-window tail stability. Provides an approximate one-step exceedance-control result and quantifies the error from next-day marking when same-contract quotes are unavailable.

Key insight

Sequential conformal recalibration on top of a base conditional-quantile forecast is a tractable way to keep VaR coverage honest on option-book tails through regime shifts, including the next-day marking-friction problem when same-contract quotes vanish.

Plain English (ELI5)

(not provided)

Tags: var, options, conformal, tail-risk, spx

2. Measuring Strategy-Decay Risk: Minimum Regime Performance and the Durability of Systematic Investing

<https://arxiv.org/abs/2026.04090>

Source: arxiv Type: regime_classifier Relevance: 0.83 Replicability: ****- (4/5)

Authors: Nolan Alexander, Frank Fabozzi

Published: 2026-04-09

TLDR

(not provided)

Summary

Introduces Minimum Regime Performance (MRP) - the lowest realized risk-adjusted return across distinct historical regimes - as a lower bound on a systematic strategy's durability. Applied to a broad universe of factor strategies, MRP exposes an efficiency-vs-resilience trade-off: high long-term Sharpe does not imply high MRP. Code on GitHub at github.com/nolanalexander/minimum_regime_performance.

Key insight

MRP turns 'strategy decay' from anecdote into a regime-conditional measurable that complements market and liquidity risk and helps rank factor strategies by robustness rather than by mean Sharpe.

Plain English (ELI5)

(not provided)

Tags: regime, factor-decay, systematic, fabozzi, code

3. Skewness Dispersion and Stock Market Returns

<https://arxiv.org/abs/2604.0091v1>

Source: arxiv Type: model Relevance: 0.80 Replicability: ***-- (3/5)

Authors: Mykola Babiak, Jozef Barunik, Josef Kurka

Published: 2026-04-09

TLDR

(not provided)

Summary

Cross-sectional dispersion in firm-level realized skewness is significantly and negatively related to future stock market returns. The signal's predictive content is incremental to a comprehensive set of 50 established predictors in both in-sample and out-of-sample tests; only six other variables beat that bar.

Key insight

Cross-sectional skewness dispersion is a previously-overlooked aggregate market-return predictor that survives a horse race against most of the predictor zoo, including in OOS tests.

Plain English (ELI5)

(not provided)

Tags: factor, skewness, predictability, barunik

4. lob-regime-scanner: HMM-based market microstructure regime detection for cryptocurrency order books

<https://github.com>

Source: github Type: regime_classifier Relevance: 0.78 Replicability: ***** (5/5)

Authors: Cameron Scarpati

Published: 2026-04

TLDR

(not provided)

Summary

End-to-end open-source pipeline that ingests Tardis L2 snapshots, computes 30+ microstructure features (multi-level OFI, VPIN, Kyle's lambda, spread/vol/autocorr), fits a 3-state Gaussian HMM (selected by BIC across $K=2..5$), decodes regimes via Viterbi, and visualizes everything in a 4-panel Plotly Dash dashboard. C++/pybind11 LOB engine handles 1M+ updates/sec; 158-test pytest suite. Reports a regime-conditional strategy with claimed Sharpe 1.8-2.5 (test set, walk-forward).

Key insight

Demonstrates a fully reproducible HMM-on-microstructure regime classifier with a working dashboard architecture - the kind of reference design to mine for OFI/VPIN feature definitions and a real-time regime overlay pattern, even if the headline Sharpe is treated skeptically (only 3 stars, no out-of-sample period beyond train/test split).

Plain English (ELI5)

(not provided)

Tags: hmm, microstructure, ofi, vpin, lob, dashboard, open-source

5. On options-driven realized volatility forecasting: Information gains via rough volatility model

<https://arxiv.org/abs/2508.10000>

Source: arxiv Type: model Relevance: 0.72 Replicability: **--- (2/5)

Authors: Zheqi Fan, Meng (Melody) Wang, Yifan Ye

Published: 2026-04-03

TLDR

(not provided)

Summary

Extracts model-based spot volatility from traded options under a rough stochastic volatility model (iterative two-step estimation a la Andersen et al. 2015a) and feeds it into the HAR family. Benchmarked against Heston, Bates, SVCJ and the VIX, the augmented HAR-RV-RHeston improves daily realized-volatility forecasting and sustains it. A deep-learning surrogate accelerates large-panel estimation.

Key insight

Spot volatility extracted from options under a rough-vol model carries forecast information that the VIX (and standard SV models) miss - a candidate input feature for HAR-style RV forecasters in a quantamental risk engine.

Plain English (ELI5)

(not provided)

Tags: volatility, options, rough-vol, har, deep-learning